

FRD Validator — Functional & Technical Specification

Document status: Draft v1.0

Prepared: 30 Jul 2026

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Scope of this document: The FRD side of the DocValidator application only. The BRD side is deferred pending an equivalent concretization pass.

Source-of-truth inputs (all under `C:\Projects\DocumentValidator`):

Input	Role in this spec
FRD Validation & Scoring Matrix.docx	The 7-category rubric, weights, 1–5 scale, quality gates — the fixed baseline.
FRD Addendum.docx	Gap analysis; enumerates every abstract term the rubric leaves undefined.
FRD Concretization Worksheet.xlsx	The machine-actionable spec: 22 criteria with required fields, mandatory/applicability flags, 1/3/5 anchors, check methods, cross-cutting decisions (A1–A7), and reference data. This is the validator's scoring config.
Functional Document Template-v4-*.docx	The canonical FRD template — the section locator's map (Sec 1–7, 4.x.y, downstream STL/CAM/NCM/GLE).

Part A — Functional Specification

A.1 Purpose

The FRD Validator ingests a Functional Requirements Document, evaluates it against the FRD Validation & Scoring Matrix (as concretized in the worksheet), and returns a defensible, weighted quality score (1.00–5.00), a quality-gate verdict, and a prioritized, evidence-cited gap report. It moves requirements-quality review from subjective manual inspection to a repeatable, auditable gate applied *before* developer handoff.

A.2 Objectives & success criteria

ID	Objective	Success measure
OBJ-1	Score any FRD authored on the v4 template against all 7 categories	22/22 criteria evaluated (or explicitly marked N/A) on a conformant FRD
OBJ-2	Produce reproducible scores	Same document + same config → identical score across runs (deterministic checks) and $\leq \pm 0.2$ variance on LLM-judged criteria across 3 runs
OBJ-3	Every score is explainable	Each criterion score cites the section/quote it derives from (decision A7)
OBJ-4	Correctly route to a gate	Verdict matches the A7 bands; near-boundary totals flagged for human review
OBJ-5	Be configurable without code changes	Anchors, weights, mandatory/applicability rules, and reference data are all data, editable by a Lead BA

A.3 Users & roles

Role	Uses the app to...
BA / Author	Self-check an FRD before submitting; act on the gap report
Lead BA / Tech Lead (Reviewer)	Run the formal gate; adjudicate human-review flags and diagram-dependent criteria; export the scorecard
SME (Config owner)	Maintain the yellow reference-data cells (sub-ledger codes, transaction types, external-system list, error codes)
Admin	Manage the config version, users, and the audit store

A.4 Assumptions, dependencies, constraints

Assumptions

- A-1: FRDs under evaluation follow the v4 Functional Document Template (Confluence "Scroll Word Exporter" .docx , or PDF export thereof).
- A-2: The concretization worksheet is authoritative for check logic; its green defaults are accepted unless a Lead BA overrides them.
- A-3: v1 assesses diagrams by **presence + caption** only (decision A5, Phase 1); diagram *content* is flagged for human review.

Dependencies

- D-1: Anthropic Claude API (model `claude-opus-5`) for LLM-judged and hybrid criteria.

- D-2: SME-supplied reference data (worksheet "yellow" cells) for Category 4/5 correctness checks. Absent this, those criteria degrade to presence checks (see A.7, BR-6).

Constraints

- C-1: The 7 categories and their published weights are fixed — the app does not add or reweight categories (per the Addendum).
- C-2: Confluence exports carry a "Scroll Word Exporter" license-warning table and use "Include Page" macros for child FPU pages; the parser must tolerate both.
- C-3: No document content is sent to any third party other than the configured Claude API for evaluation; PII handling follows existing DocEmbedder guardrails.

A.5 Functional requirements

Ingestion & structure

- **FR-1 — Document upload.** Accept `.docx` and `.pdf` FRDs (single file; child FPU pages either inlined by the export or uploaded as a set).
- **FR-2 — Parsing.** Extract headings (with numbering), paragraphs, tables (as row/cell matrices), and embedded images with their captions/alt-text and position. Preserve section hierarchy.
- **FR-3 — Noise stripping.** Discard the Scroll Word Exporter license-warning table and other export boilerplate before evaluation.
- **FR-4 — Section location.** Map extracted content to canonical template sections (Sec 1–7 and sub-sections 3.1–3.5, 4.1–4.6, 4.1.4/4.3.7/4.5.9/4.6.9, NFR, Sec 5 AC, Sec 6 Open Questions, Sec 7 References) by heading text + numbering, with an LLM fallback classifier when numbering is absent (decision A3).
- **FR-5 — FPU detection.** Identify each Functional Process Unit, its type (UI / Report / Batch / Dashboard / Sync-Interface / Async-Interface), and its ID; associate each with its Sec 4.x subsection (decision/reference: FPU definition + `FPU-<TYPE>-<nnn>` scheme).

Applicability

- **FR-6 — Applicability resolution.** For each Conditional criterion, evaluate its applicability rule against detected FPUs/interfaces/components and mark it **applicable** or **N/A** (e.g. async strategies apply only if an async interface exists; downstream components apply per component the transaction hits). Reference BR-2.

Evaluation

- **FR-7 — Criterion evaluation.** Evaluate all 22 criteria (Appendix D) using the criterion's configured check method:
 - **Rule** — deterministic presence/structure/arithmetic check.
 - **LLM** — qualitative judgment via Claude against the criterion's 1/3/5 anchors.
 - **Hybrid** — rule for presence/structure + LLM for quality; combine per BR-4.
- **FR-8 — Required-fields check.** For each criterion, verify presence of its "Required Fields / Elements" (worksheet column) — e.g. version table {version, date, author, change desc, approver}; payload per-field {name, type, required, constraints, description}; STL {SSI, dates, currency, amount...}.

- **FR-9 — Correctness check (config-gated).** Where reference data exists, validate values (e.g. sub-ledger code $\in$ code set; interface protocol $\in$ allowed set; Dr total = Cr total for sample movements). Where reference data is absent, fall back to presence-only and annotate the finding (BR-6).
- **FR-10 — Diagram handling.** For diagram-dependent criteria (BPD, OFD swim-lanes, flowcharts, UI mockups, sequence diagrams), verify image presence and caption/label in the expected section; mark the criterion **human-review-required** for content assessment in v1 (A5 Phase 1). (v2: vision-model content check — see roadmap.)
- **FR-11 — Evidence capture.** For every criterion score, capture the section reference and a supporting quote/snippet (or "not found") (decision A7).

Scoring & gating

- **FR-12 — Bullet→category combine.** Combine each category's criterion (bullet) scores into a single 1–5 using BR-1 (average → nearest 0.5, with the floor rule).
- **FR-13 — N/A renormalization.** Exclude N/A criteria and renormalize remaining weights per BR-2.
- **FR-14 — Weighted total.** Compute the weighted total across categories, rounded to 2 dp (BR-3 / A7).
- **FR-15 — Gate verdict.** Map the total to a gate band (BR-3): ≥ 4.50 Ready; 3.50–4.49 Conditional; < 3.50 Return to BA.
- **FR-16 — Boundary flag.** Any total within 0.10 of a band boundary (3.50 / 4.50) is flagged **mandatory human review** (A7).

Output

- **FR-17 — Scorecard.** Present overall score, gate verdict, per-category scores, and per-criterion scores with evidence and N/A markers.
- **FR-18 — Gap report.** List failing/low criteria ranked by leverage (weight $\times$ shortfall), each with the concrete missing element and the anchor language explaining the score.
- **FR-19 — Human-review queue.** Surface all criteria flagged for human review (diagram-content, near-boundary) for reviewer adjudication; a reviewer can override a criterion score with a mandatory justification.
- **FR-20 — Export.** Export the scorecard + gap report to PDF/Word with the audit trail (aligns with existing DocValidator/IKA export patterns).

Configuration & audit

- **FR-21 — Config load.** Load the scoring config (criteria, anchors, mandatory/applicability, weights, check methods) and reference data from a versioned config artifact derived from the worksheet.
- **FR-22 — Config edit.** Allow a Lead BA to review/edit green defaults and an SME to supply yellow reference data through an admin surface, producing a new config version.
- **FR-23 — Run audit.** Persist every evaluation run: document identity/hash, config version, per-criterion scores + evidence, overrides, final score, verdict, timestamp, and Claude usage/cost.

A.6 Business rules (the scoring model — decisions A1, A4, A7)

- **BR-1 (A1) — Category combine.** Category score = arithmetic mean of its criterion scores, rounded to nearest 0.5. **Floor rule:** if any criterion in the category scored **1** (a critical element missing), cap the

category at 2.

- **BR-2 (A4) — Applicability & renormalization.** A Conditional criterion whose trigger is genuinely absent scores **N/A**, is excluded, and the category's remaining criteria are re-weighted to 100% of the category. A category that is entirely N/A redistributes its weight proportionally across the remaining categories.
- **BR-3 (A7) — Gate mechanics.** Weighted total rounded to 2 dp. Bands inclusive at the lower bound: **≥ 4.50** Ready for Sprint / Sign-Off; **3.50–4.49** Conditional Pass with Rework; **< 3.50** Gate Failure / Return to BA. Any total within **0.10** of 3.50 or 4.50 → mandatory human review.
- **BR-4 — Hybrid combine.** For a Hybrid criterion, a failed deterministic presence check caps the criterion score at the "missing/partial" anchor (≤ 3) regardless of LLM judgment; the LLM judgment sets the score within the range the presence check allows.
- **BR-5 — Anchors.** Each criterion is scored on 1/3/5 anchors (Appendix D); 2 and 4 interpolate. 1 = Unacceptable, 3 = Rework, 5 = Production Ready.
- **BR-6 — Presence-vs-correctness degradation.** When correctness reference data is unavailable for a criterion, the check runs in presence-only mode and the finding is annotated "presence verified; correctness not checked (reference data pending)".

A.7 Non-functional requirements

ID	Requirement
NFR-1 (Performance)	A full 7-category evaluation of a typical FRD completes within ~2–3 min (LLM-bound); support async/batch for large docs.
NFR-2 (Reproducibility)	Deterministic criteria are byte-stable; LLM criteria pinned to <code>claude-opus-5</code> with fixed prompts and structured outputs; runs record model + config version.
NFR-3 (Auditability)	Every score traces to config version + evidence; every override records actor + justification.
NFR-4 (Extensibility)	Adding/retuning a criterion is a config edit, not a code change.
NFR-5 (Security/Privacy)	Document content leaves the environment only to the configured Claude API; PII guardrails applied pre-send; secrets in env/secret store.
NFR-6 (Cost transparency)	Claude token usage and cost recorded per run and shown to admins (aligns with existing cost-tracking).
NFR-7 (Usability)	Reviewer can go from upload to adjudicated verdict without leaving the app.

A.8 Acceptance criteria (for the validator itself)

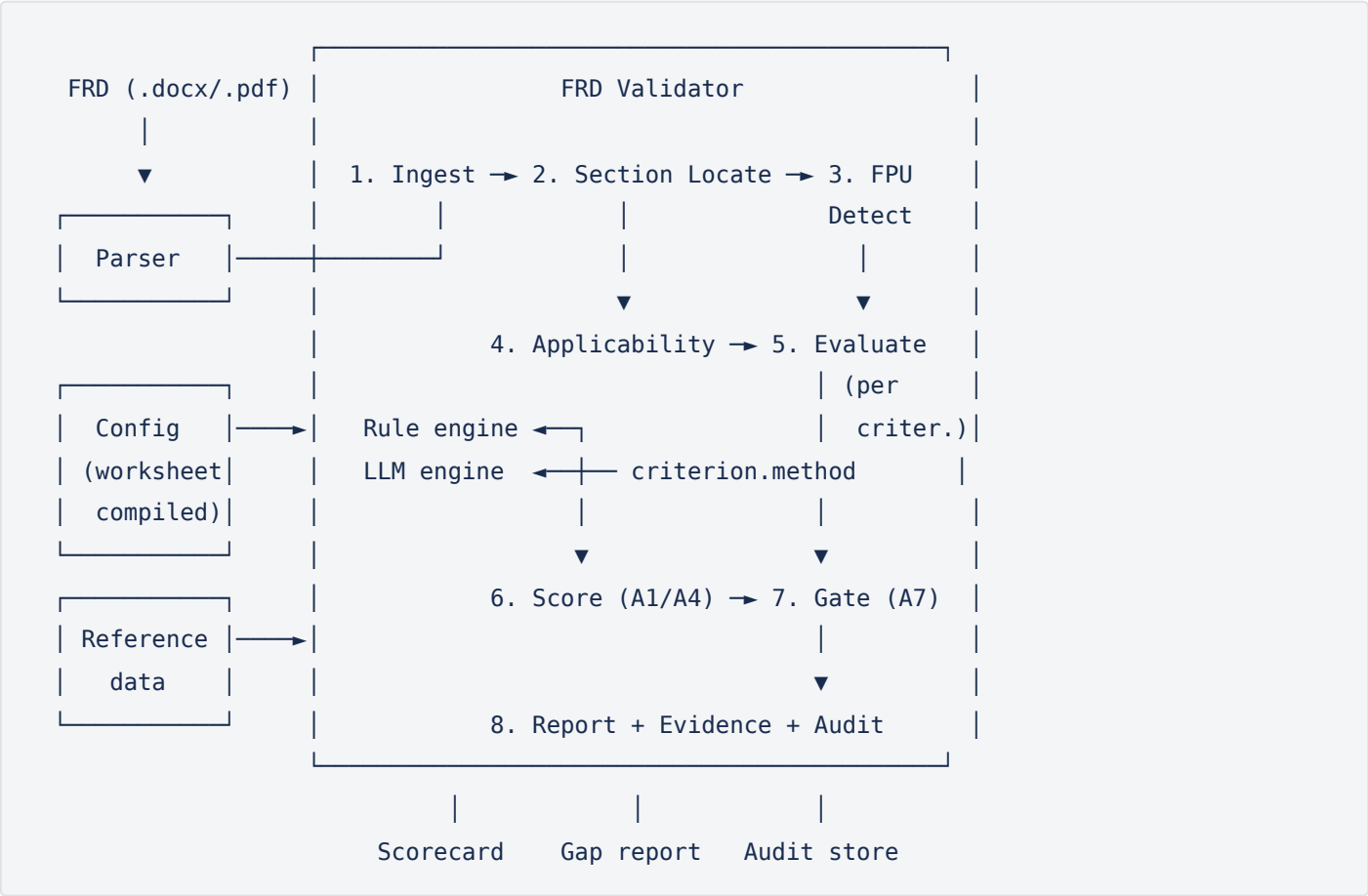
- AC-1: On a fully-populated sample FRD, every applicable criterion returns a 1–5 score with non-empty evidence; every Conditional-absent criterion returns N/A.
- AC-2: The weighted total, category scores, N/A renormalization, and gate verdict reproduce a hand-worked example within rounding.
- AC-3: A doc with a missing critical element (a category criterion = 1) shows that category capped at 2 (BR-1).

- AC-4: A doc scoring 3.48 and one scoring 4.52 are both flagged for mandatory human review (BR-3 boundary).
- AC-5: Diagram-dependent criteria on a doc with images present but no reference data show "presence verified, human-review-required".
- AC-6: Editing an anchor/weight/reference-value in config changes the next run's result with no code deployment.

Part B — Technical Specification

B.1 Architecture overview

A hybrid **rules + LLM** pipeline. Deterministic checks run locally; qualitative checks call Claude. The worksheet is compiled into a machine-readable config that drives the engine.



B.2 Technology stack

Chosen to match the existing DocEmbedder/DocValidator ecosystem (Python + Streamlit + Claude + SQLite).

Layer	Choice	Notes
Language	Python 3.11+	Matches existing projects

Layer	Choice	Notes
UI	Streamlit	Reuse app shell, auth gate, export, cost-tracking, theme from DocValidator
DOCX parsing	<code>python-docx</code>	Already available; reads tables/headings; images via the <code>.docx</code> zip parts
PDF parsing	<code>pypdf</code> / <code>pdfplumber</code>	For PDF-exported FRDs; tables via <code>pdfplumber</code>
LLM	<code>anthropic</code> SDK, model <code>claude-opus-5</code>	Adaptive thinking; structured outputs via <code>output_config.format</code> ; prompt caching for the shared config/system prompt; optional Batch API for cost
Config	YAML/JSON compiled from the worksheet	One file, version-controlled
Persistence	SQLite	Runs, scores, evidence, overrides, cost — matches existing SQLite usage
Export	<code>python-docx</code> / <code>reportlab</code>	Reuse IKA-branded export

The LLM layer uses the official Anthropic SDK and `claude-opus-5`. Structured outputs (`messages.parse()` / `output_config.format`) guarantee schema-valid per-criterion results; prompt caching on the (large, stable) config+rubric system prompt cuts cost across the 22 criteria; the Batch API is available for non-interactive bulk scoring at 50% cost.

B.3 Component design

Component	Responsibility	Key I/O
Ingestor	Parse <code>.docx</code> / <code>.pdf</code> → structured <code>Document</code> (sections, tables, images, captions). Strip export noise (C-2).	<code>file</code> → <code>Document</code>
Section Locator	Map <code>Document</code> blocks → canonical template sections; numbering-first, LLM-classifier fallback (A3).	<code>Document</code> → <code>SectionMap</code>
FPU Detector	Find FPUs, types, IDs; associate to Sec 4.x.	<code>SectionMap</code> → <code>List[FPU]</code>
Applicability Resolver	Evaluate each Conditional criterion's applicability rule vs FPUs/interfaces/components (A4).	<code>criteria + FPUs</code> → <code>{criterion_id: applicable / N/A}</code>
Rule Engine	Deterministic checks: field presence, list non-empty, enum membership, Dr=Cr balance, link reachability.	<code>criterion + evidence spans</code> → <code>RuleResult</code>

Component	Responsibility	Key I/O
LLM Engine	Claude call per LLM/Hybrid criterion; structured output {score, evidence, rationale} against anchors.	criterion + section text → <code>LLMResult</code>
Scorer	Combine bullets (A1), renormalize N/A (A4), weight, round (A7).	criterion scores → category + total
Gate	Band mapping + boundary flag (A7).	total → verdict + flags
Reporter	Scorecard, ranked gap report, human-review queue, export.	results → UI + files
Config Service	Load/edit/version the compiled worksheet config + reference data.	worksheet → <code>Config</code>
Audit Store	Persist runs, scores, evidence, overrides, cost.	run → SQLite

B.4 Data models

Config (compiled from the worksheet) — one entry per criterion:

```
{
  "id": "5.2",
  "category": "5. Integration Contract & Payload Quality",
  "category_weight": 0.15,
  "template_sections": ["4.5", "4.6"],
  "criterion": "Schema Definition",
  "required_fields": ["request payload", "response payload",
    "per-field: name", "data type", "required/optional",
    "constraints", "description", "error codes"],
  "mandatory": "conditional",
  "applicability_rule": "per interface present",
  "anchors": {
    "1": "No payload schemas.",
    "3": "Payloads listed but missing data types, requirement rules, or response/error codes.",
    "5": "Request & response fully itemized with strict types, requirement rules, descriptions,
  },
  "reference_values_ref": "error_code_catalog",
  "check_method": "hybrid"
}
```

Cross-cutting decisions (A1–A7) and reference data (external systems, sub-ledger codes, balance/transaction/account types, interface types/protocols/formats, error-code catalog, NFR dimensions, async thresholds, idempotency pattern, edge-case catalog, AC format) are stored as first-class config sections, each tagged `source: green|yellow` so the UI can show what still needs an SME.

Evaluation result (per criterion):

```
{
  "criterion_id": "5.2",
  "status": "scored",          // scored | n/a | human_review
  "score": 4,                  // 1-5, or null if n/a
  "check_method": "hybrid",
  "presence": {"passed": true, "missing": []},
  "correctness": {"checked": false, "reason": "error_code_catalog pending (yellow)"},
  "evidence": [{"section": "4.5", "quote": "Request payload table: field 'tradeId' string requi"},
  "rationale": "Request & response itemized with types; error codes not enumerated → anchor 3-4",
  "flags": ["correctness_not_checked"]
}
```

Run record: {run_id, doc_name, doc_sha256, config_version, results[], category_scores, total, verdict, human_review_items[], overrides[], usage{tokens, cost}, created_at, actor}.

B.5 Section locator (A3) — algorithm

1. Build the canonical section tree from the template (headings + numbers).
1. For each document heading: normalize text; match to canonical by (a) exact section number, else (b) heading-text similarity $\geq$ threshold.
1. Unmatched or unnumbered blocks (common in Confluence child-page includes): batch to an LLM classifier that assigns each to the most likely canonical section (or "unknown").
1. Produce `SectionMap`: `canonical_section` → `[content blocks]`. Sections with no content are recorded as **absent** (feeds presence checks and applicability).

B.6 FPU detection

- Parse the "Requirement Detail" FPU list and each `Functional Requirement# N` heading.
- Classify FPU type from the subsection shape (UI has Screen Components/Validation; Report has the report summary matrix; Batch has Execution Schedule; Dashboard has the feed summary matrix; Sync/Async have the interfacing summary matrix + payloads).
- Assign/verify `FPU-<TYPE>-<nnn>` IDs; associate each FPU to its Sec 4.x subsection. FPU set drives applicability (which of Cat 3 sub-types, Cat 4 components, Cat 5 sync/async apply).

B.7 Criterion evaluation engine

```
for criterion in config.criteria:
    if criterion.mandatory == "conditional" and not applicable(criterion, fpus):
        result = N/A
        continue
```

```

spans = section_map.text_for(criterion.template_sections, fpu_scope)
if criterion.check_method in ("rule", "hybrid"):
    rule = run_rule_checks(criterion, spans, reference_data)    # presence/structure/arith
if criterion.check_method in ("llm", "hybrid"):
    llm = claude_score(criterion, spans)                        # structured {score,evidence}
result = combine(criterion, rule, llm)                          # BR-4 for hybrid
if criterion.is_diagram_dependent and phase == 1:
    result.flag("human_review")                                # A5 Phase 1

```

Representative rule checks: version-table field presence (1.1); In/Out-scope bullets non-empty (2.1); FPU decomposition + unique IDs (3.1); Dr total = Cr total on sample movements (4.3); interface {name,type,protocol,format} present and $\in$ allowed sets (5.1); request+response payload per-field completeness (5.2); async retry/throttle/rate-limit present when async exists (5.5); NFR numeric-target presence (6.2); ≥ 1 happy $\pm \geq 1$ edge scenario (7.1); AC in Given/When/Then (7.2).

B.8 LLM integration

- **Model:** `claude-opus-5` (default; do not downgrade without instruction).
- **One call per LLM/Hybrid criterion**, or batched per category. System prompt carries the rubric + that criterion's 1/3/5 anchors + required fields; user content carries the located section text (and table text). **Prompt caching** on the stable system/config prefix across the 22 criteria.
- **Structured output** via `output_config.format` (JSON schema): `{score:int 1-5, evidence:[{section,quote}], rationale:string, missing:[string]}` — guarantees parseable, schema-valid results.
- **Adaptive thinking** on `(thinking:{type:"adaptive"})` at `effort:"high"` for judgment quality.
- **Determinism:** temperature is not a knob on this model; reproducibility comes from fixed prompts + pinned model + structured output; report ± 0.2 tolerance (OBJ-2).
- **Cost:** record `usage` per call; offer **Batch API** (50% cost) for non-interactive bulk runs. Cite `claude-opus-5` pricing \$5/\$25 per MTok for estimates.
- **Guardrails:** check `stop_reason == "refusal"` before reading content; apply PII pre-filter before send (NFR-5).

B.9 Scoring algorithm (A1/A4/A7)

```

# category combine (BR-1 / A1)
def category_score(scores):
    if not scores: return None
    avg = round_to_half(mean(scores))
    if any(s == 1 for s in scores): avg = min(avg, 2)
    return avg

# weight renormalization (BR-2 / A4)
active = [c for c in categories if c.score is not None]
tw = sum(c.weight for c in active)
for c in active: c.norm_weight = c.weight / tw

```

```
# weighted total + gate (BR-3 / A7)
total = round(sum(c.score * c.norm_weight for c in active), 2)
verdict = "READY" if total >= 4.50 else "CONDITIONAL" if total >= 3.50 else "RETURN"
if min(abs(total-3.50), abs(total-4.50)) <= 0.10: flag_human_review(total)
```

B.10 Evidence, reporting & human-in-the-loop

- Each criterion result stores section + quote (A7). The gap report ranks findings by **leverage = category_norm_weight × (5 – score)** so reviewers see the highest-impact gaps first.
- The **human-review queue** holds diagram-content criteria (A5 Phase 1) and near-boundary runs (A7). Reviewer overrides require a justification and are audited; overrides recompute the total.
- Export mirrors the existing IKA-branded PDF/Word output.

B.11 Persistence

SQLite tables: `runs`, `criterion_results`, `evidence`, `overrides`, `config_versions`, `reference_data`, `usage_cost`. The audit store makes any historical verdict reproducible against its config version.

B.12 Error handling

- Parse failure → surface which section/table failed; do not silently truncate.
- Claude refusal / rate-limit / timeout → retry with backoff (SDK default) then mark the criterion `evaluation_error` (never a silent 0).
- Missing reference data → presence-only degrade (BR-6), not a hard failure.
- Section not found → treated as **absent** (presence fail), which is a legitimate low score, not an error.

B.13 Testing strategy

- **Unit:** each rule check; the scorer (BR-1/2/3 incl. floor rule, full-N/A redistribution, boundary flags) against hand-worked examples (AC-2/3/4).
- **Golden FRDs:** a small labelled set (fully-populated, gap-ridden, async-only, no-downstream) → assert expected category/total/verdict.
- **LLM harness:** re-run judged criteria 3× → assert $\leq \pm 0.2$ variance; snapshot prompts.
- **Config round-trip:** worksheet → compiled config → 22 criteria present with anchors, methods, weights summing to 100%.

B.14 Phasing / roadmap

Phase	Scope
v1 (build-ready now)	Full pipeline; all 22 criteria; rule+LLM+hybrid; A1/A4/A7 scoring; presence-first (correctness where green reference data exists); diagrams = presence + human-review flag ; scorecard + gap report + export + audit.

Phase	Scope
v1.1	Populate yellow reference data (sub-ledger codes, transaction types, external-system list, error codes) → turn on Cat 4/5 correctness checks.
v2	Vision-model diagram content assessment (BPD/OFD/flowcharts/mockups/sequence) via <code>claude-opus-5</code> image input; link reachability HTTP checks (1.3); calibrate against a scored corpus.
v3	BRD Validator (after a BRD concretization worksheet + template exist), sharing the same engine.

B.15 Risks & mitigations

Risk	Mitigation
Confluence exports vary (child-page includes, macros) → section locator misses	Numbering-first + LLM fallback (A3); mark unresolved sections "unknown" for review, never silent
Diagram-heavy criteria under-automated in v1	Explicit A5 Phase-1 policy: presence + human-review flag; v2 vision
Correctness checks blocked on SME data	BR-6 presence-only degrade keeps v1 useful; v1.1 lights them up
No labelled corpus → LLM scores untrusted early	Treat scores advisory; A7 boundary + diagram flags force human review; build corpus in v2
LLM cost on 22-criteria runs	Prompt caching on the shared prefix; Batch API for bulk; per-run cost tracking

B.16 Open items — SME inputs required (worksheet "yellow" cells)

Non-blocking for v1 (BR-6), required for full correctness in v1.1:

- Sub-ledger / GL code set (4.1, 4.2)
- Transaction types (4.2)
- Full external-system list beyond I-STAR / GV (2.3)
- Actual error-code values (5.2, 6.1)
- Exact SSI attribute set/format confirmation (4.1, 4.3)
- Confirmations of proposed green defaults: approver roles (1.1), balance/account types (4.2/4.3), interface type/protocol/format closed sets (5.1), NFR dimensions (6.2), idempotency key + window (6.3), AC format (7.2).

Appendix C — Cross-cutting decisions (A1–A7), as implemented

#	Decision	Implementation
A1	Category combine	BR-1: mean → nearest 0.5, any-1 caps category at 2
A2	Score anchors	Adopt worksheet 1/3/5 anchors (Appendix D); 2/4 interpolate
A3	Template + section locator	B.5: numbering-first + LLM fallback against v4 template
A4	Applicability / renormalization	FR-6 + BR-2
A5	Visual-artifact policy	v1 presence + human-review flag; v2 vision (B.14)
A6	Domain reference data	Config <code>reference_data</code> section; green pre-filled, yellow SME-pending (B.16)
A7	Evidence + gate mechanics	FR-11 + BR-3: evidence citations, 2-dp rounding, inclusive bands, 0.10 boundary flag

Appendix D — Criteria catalog (22 criteria)

Weights: Cat1 10% · Cat2 15% · Cat3 20% · Cat4 20% · Cat5 15% · Cat6 10% · Cat7 10%.

Mandatory: **Y** = always; **C** = conditional (applicability rule). Method: R=rule, L=LLM, H=hybrid.

ID	Category	Criterion	Mand.	Method	Applicability
1.1	1 Governance & Base Readiness	Document Control	Y	H	Always
1.2	1	Contextual Alignment	Y	L	Always
1.3	1	Dependency Resolution	Y	H	Always
2.1	2 Core Scoping & Topography	Granular Scoping	Y	H	Always
2.2	2	Process Modeling (BPD)	Y	H (diagram)	Always
2.3	2	Topography & Integration (OFD)	C	H (diagram)	External-system integration present
3.1	3 FPU Specification	Modular Breakdown	Y	H	Always
3.2	3	Interface & Field Schemas	C	H (diagram)	Per FPU types present

ID	Category	Criterion	Mand.	Method	Applicability
3.3	3	Processing Logics	Y	H	Always (content varies by FPU)
4.1	4 Downstream & Financial	Sub-Component Processing (STL/CAM/NCM/GLE)	C	H	Per component the txn hits
4.2	4	Sub-Ledger & Posting Rules	C	H	When it posts to the ledger
4.3	4	State & Sample Data Validation	C	H (rule: Dr=Cr)	When the component produces movements
5.1	5 Integration Contract & Payload	Interface Contracts	C	R+L	Sync/async interfaces exist
5.2	5	Schema Definition	C	H	Per interface present
5.3	5	Transformation Logic	C	H (diagram)	Per interface present
5.4	5	Sequence Workflow Diagram	C	H (diagram)	Interfaces exist
5.5	5	Async Processing Strategies	C	H	Async interface exists
6.1	6 Resilience & NFR	Error & Exception Management	Y	H	Always
6.2	6	Core NFR Coverage	Y	H	Always
6.3	6	Lifecycle Operations (idempotency/back-date/migration)	C	H	Where txns apply / migration relevant
7.1	7 Verifiability	Concrete Scenarios	Y	H	Always
7.2	7	Acceptance Criteria	Y	H	Always

(Full required-fields, 1/3/5 anchors, and reference values for each criterion live in the compiled config, sourced verbatim from the FRD Concretization Worksheet.)